# ALEX BROUGHTON

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## EDUCATION AND ACADEMIC APPOINTMENTS

**SLAC National Accelerator Laboratory, Stanford University, KIPAC** *2024 – present*
Postdoctoral Research Associate

**University of California, Irvine** *2024*
Ph.D., Physics
*Thesis Title:* Advanced Pixel-Level Analyses for Large-Scale Surveys of the Dark Universe

**University of California, Davis** *2019*
B.S., Physics

## RESEARCH INTERESTS

Instrument signature removal, image calibration, CCD sensor systematics, weak lensing and cosmic shear, survey design, star-galaxy separation, cross-survey joint processing, galaxy clusters, AI/ML methods for astrophysics

## PUBLICATIONS

**Broughton**, **Alexander**, Agnès Ferté, Christopher Z. Waters, Eli S. Rykoff, Sean Patrick MacBride, Andrés A. Plazas Malagón, Robert H. Lupton, and Elana K. Urbach. *RTN-117: Image Calibration and Instrument Signal Removal for the First Year of Rubin LSSTCam*. Technical Note. Rubin Category 2 Key Paper (in prep.) NSF-DOE Vera C. Rubin Observatory, 2026. https://rtn-117.lsst.io.

Gomez-Jimenez, Manuel, Craig Lage, **Alex Broughton**, and 24 colleagues. "Vera C. Rubin's auxiliary telescope ascent: fault mitigation and the transition from construction to reliable operations". In: *Observatory Operations: Strategies, Processes, and Systems XI*. Vol. 14151. Proc. SPIE. Aug. 2026. DOI: 10.1117/12.3104984.

Han, Jiwon Jesse, Anirudh Chiti, Kai-Feng Chen, and 93 colleagues, incl. **Alex Broughton**. "A Path to an All-Sky Survey with Roman". In: *arXiv e-prints*, arXiv:2602.21280 (Feb. 2026), arXiv:2602.21280. DOI: 10.48550/arXiv.2602.21280.

MacBride, Sean Patrick, Aaron Roodman, Stuart Marshall, and 88 colleagues, incl. **Alex Broughton**. "The On-Sky Performance of the LSST Camera CCD Array". In: *X-Ray, Optical, and Infrared Detectors for Astronomy XII*. Vol. 14157. Proc. SPIE. Aug. 2026, 141570Y. DOI: 10.1117/12.3102906.

Rodeghiero, Gabriele, Alex Drlica-Wagner, Alessio Taranto, and 78 colleagues, incl. **Alex Broughton**. "An overview of the stray light findings and interpretation during the on-sky commissioning of LSSTCam". In: *Ground-based and Airborne Telescopes XI*. Vol. 14147. Proc. SPIE. Aug. 2026. DOI: 10.1117/12.3103019.

Vera C. Rubin Observatory Team. *RTN-115: The Vera C. Rubin Observatory Data Preview 2*. Technical Note. Co-lead, Data Release Processing section; **A. Broughton** (submitted). NSF-DOE Vera C. Rubin Observatory, 2026. https://rtn-115.lsst.io.

OpenUniverse, A. Alarcon, L. Aldoroty, and 55 colleagues, incl. **Alex Broughton**. "OpenUniverse2024: a shared, simulated view of the sky for the next generation of cosmological surveys". In: *Monthly Notices of the Royal Astronomical Society* 544.4 (Oct. 2025), pp. 3799–3823. DOI: 10.1093/mnras/staf1833.

Plazas Malagón, Andrés A., Seth W. Digel, Aaron Roodman, **Alex Broughton**, and LSST Camera Team. *CTN-001: LSSTCam and LSSTComCam Focal Plane Layouts*. Technical Note. NSF-DOE Vera C. Rubin Observatory, 2025. DOI: 10.71929/RUBIN/2584019.

Plazas Malagón, Andrés A., Chris Waters, **Alex Broughton**, Eli Rykoff, Agnès Ferté, Merlin Fisher-Levine, and Robert Lupton. "Instrument Signature Removal and Calibration Products for the Rubin Legacy Survey of

Space and Time". In: *Journal of Astronomical Telescopes, Instruments, and Systems* 11.1, 011209 (Feb. 2025), p. 011209. DOI: 10.1117/1.JATIS.11.1.011209.

Rubin Observatory Science Pipelines Developers, Yusra AlSayyad, Tim S. Axelrod, and 77 colleagues, incl. **Alex Broughton**. *PSTN-019: The LSST Science Pipelines Software: Optical Survey Pipeline Reduction and Analysis Environment*. Technical Note. (submitted). NSF-DOE Vera C. Rubin Observatory, 2025. DOI: 10.71929/RUBIN/2570545.

Vera C. Rubin Observatory Team. *RTN-095: The Vera C. Rubin Observatory Data Preview 1*. Technical Note. 331 authors, including **A. Broughton** (submitted). NSF-DOE Vera C. Rubin Observatory, 2025. DOI: 10.71929/RUBIN/2570536.

**Broughton**, **Alex**, Yousuke Utsumi, Andrés A. Plazas Malagón, Christopher Waters, Craig Lage, Adam Snyder, Andrew Rasmussen, Stuart Marshall, Jim Chiang, Simona Murgia, and Aaron Roodman. "Mitigation of the Brighter-fatter Effect in the LSST Camera". In: *Publications of the Astronomical Society of the Pacific* 136.4 (Apr. 2024), p. 045003. DOI: 10.1088/1538-3873/ad3aa2.

Roodman, A., A. Rasmussen, A. Bradshaw, and 73 colleagues, incl. **Alex Broughton**. "LSST camera verification testing and characterization". In: *Ground-based and Airborne Instrumentation for Astronomy X*. Ed. by Julia J. Bryant, Kentaro Motohara, and Joël. R. D. Vernet. Vol. 13096. Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series. July 2024, 130961S, 130961S. DOI: 10.1117/12.3019698.

Utsumi, Yousuke, Pierre Antilogus, Pierre Astier, and 30 colleagues, incl. **Alex Broughton**. "LSST Camera focal plane optimization". In: *X-Ray, Optical, and Infrared Detectors for Astronomy XI*. Ed. by Andrew D. Holland and Kyriaki Minoglou. Vol. 13103. Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series. Aug. 2024, 131030W, 131030W. DOI: 10.1117/12.3019117.

Esteves, Johnny H., Yousuke Utsumi, Adam Snyder, and 14 colleagues, incl. **Alex Broughton**. "Photometry, Centroid and Point-spread Function Measurements in the LSST Camera Focal Plane Using Artificial Stars". In: *Publications of the Astronomical Society of the Pacific* 135.1053 (Nov. 2023), p. 115003. DOI: 10.1088/1538-3873/ad0a73.

Karwin, Christopher M., **Broughton, Alex**, Simona Murgia, Alexander Shmakov, Mohammadamin Tavakoli, and Pierre Baldi. "Improved modeling of the discrete component of the Galactic interstellar $\gamma$-ray emission and implications for the Fermi–LAT Galactic Center excess". In: *Phys. Rev. D* 107 (12 June 2023), p. 123032. DOI: 10.1103/PhysRevD.107.123032.

Shmakov, Alexander, Mohammadamin Tavakoli, Pierre Baldi, Christopher M. Karwin, **Alex Broughton**, and Simona Murgia. "Deep learning models of the discrete component of the Galactic interstellar $\gamma$-ray emission". In: *Phys. Rev. D* 107 (6 Mar. 2023), p. 063018. DOI: 10.1103/PhysRevD.107.063018.

Brown, Richard, Stephen Czarnecki, Michael Spears, Bruce Nordman, **Alex Broughton**, Michael Von Hippel, and Anna Liao. *Price Based Local Power Distribution Management System (Local Power Distribution Manager) v1.0*. [Computer Software] https://doi.org/10.11578/dc.20210416.76. Nov. 2016. DOI: 10.11578/dc.20210416.76.

## HONORS & AWARDS

**KIPAC Innovation Grant ($15,000)** *2025*
Kavli Institute for Particle Astrophysics and Cosmology (KIPAC), Stanford University

**Graduate Student Research (SCGSR) Fellowship** *2022*
Department of Energy Office of Science (DOE)

**Data Science Fellowship (DSF)** *2021*
Physical Sciences Machine Learning (PSML) NEXUS, UC Irvine

## CONFERENCE PRESENTATIONS & INVITED TALKS

**A new survey, optimized to test ΛCDM with Rubin in 2036** *Sept. 2026*
KAS 2026, Cambridge

**Fast, Physics-Driven Data Production at Petabyte-Scale: Making LSST Data AI-Ready** *Sept. 2026*
KAS 2026, Cambridge

**Physics-Driven Data Production at Petabyte Scale, and the Science It Enables for Rubin 2.0** *Sept. 2026*
Laboratoire de Physique Nucléaire et de Hautes Énergies (LPNHE)

**Rubin 2036 Workshop Recap: What would you do with a discovery machine?** *Aug. 2026*
KIPAC Tea, SLAC National Lab/Stanford University

**Fast, Physics-Driven Data Production at Petabyte-Scale: A Preview of the First LSST Data** *Aug. 2026*
C4DU @ CoDA

**Lightning Talk: Brighter-Fatter Effect in LSSTCam** *June 2026*
JWST Focus Group on the Brighter-Fatter Effect

**Catching the First Photons with the Vera C. Rubin Observatory** *March 2026*
Universidad Andrés Bello (UNAB), Chile

**Catching the First Photons with the Vera C. Rubin Observatory** *July 2025*
Pontificia Universidad Católica de Valparaíso (PUCV)

**Catching the First Photons with the Vera C. Rubin Observatory** *June 2025*
Universidad Técnica Federico Santa María (USM)

**Wait... where does my data come from?!** *July 2024*
Rubin Observatory Undergraduate Summer Workshop

**Constraining Astrophysical Parameters with CCD Systematics** *March 2024*
Image Sensors for Precision Astronomy (ISPA), SLAC National Lab

**Instrument-to-Science: Enabling the Legacy Survey of Space and Time (LSST)** *Jan. 2024*
AstroLunch Seminar Series, University of Washington (UW)

**Pixels-to-Science: Enabling the Legacy Survey of Space and Time** *Jan. 2024*
Rubin@SLAC Seminar Series, SLAC National Lab/Stanford University

**Instrument-to-Science: Enabling the Legacy Survey of Space and Time (LSST)** *2023*
Colloquium, CalPoly SLO

**The Brighter-Fatter Effect in LSSTCam: Correction and Influence on Science** *2023*
KIPAC Rubin Science Talk, Stanford University

**Impact of the BFE on Cosmic Shear** *2023*
Dark Energy Science Collaboration (DESC) Meeting, SLAC National Lab

**Modeling $\gamma$-Ray Emission with Deep Learning and Implications for Dark Matter** *2021*
PSML Nexus Seminar Series, Irvine, CA

**Bias Persistence in Run 4 LSST Camera Testing** *2021*
Dark Energy Science Collaboration (DESC) Meeting, Virtual

**Characterizing Newly Discovered ITL Sensor 'Comb' Pattern** *2020*
Dark Energy Science Collaboration Meeting, Virtual

**Simulation of Light Yield and Collection Efficiency in a Liquid Xenon Detector** *2018*
UC Davis Undergraduate Research Conference, Davis, CA

## OUTREACH & LEADERSHIP

**Rubin 2036 – SOC/LOC Organizer** *2026*

Workshop dedicated to planning future survey and instrumentation for the Simonyi Telescope post-LSST's 10-year survey.

**KIPAC Seminar Organizer** *2026 – 2027*

Organize the KIPAC Seminar series for the 2026–2027 academic year.

**La Serena Rubin Science Workshop – Lead Organizer** *2025*

Workshop dedicated to early Rubin Science and synergies with other NSF/NOIRLab/AURA telescopes, bringing in project team members and local Universidad de La Serena colleagues.

**Co-organizer, SLAC Fundamental Physics Directorate Experimental Seminar** *2025 – present*

Invite and host speakers in observational and experimental cosmology and astrophysics for the SLAC FPD Experimental Seminar series.

**KIPAC VisLab Lead** *2024 – present*

Lead the KIPAC Visualization Lab, presenting immersive 3D visualizations of the Universe to visitors and prospective donors at SLAC.

**Co-founder, KIPAC-Chile Engagement Program** *2024 – present*

The KIPAC-Chile Engagement Program, launched through a KIPAC Innovation Grant, gives KIPAC members who travel to Chile access to specialized funds meant to help develop professional and personal connections, and spark new research collaborations with scientists at institutions across Chile. Managed a budget of $15,000 and organized the travel of 10 institutional researchers to give talks at all universities in Chile with PhD programs in astronomy/physics. Developed new metrics to track engagement and impact after 2 years.

**Save The Bay (Non-Profit)** *2023 – present*

Permanent active member for political advocacy, education, restoration, and conservation of native wildlife and habitat in the San Francisco Bay Area.

## TEACHING/MENTORSHIP EXPERIENCE

**Guest Lecturer (2 lectures)** *Fall 2026*
PHYS 367: Observational Data Analysis, Stanford University

**Mentor, Undergraduate Research Student** *Summer 2026*
Don Wong, Brown University: star-galaxy separation, KIPAC

**Mentor, Graduate Research Student**
Jamar Kittling, Stanford University: photometric calibration for LSST

**Teaching Assistant** *Fall 2019*
PHYSICS 2: Introduction to Mathematical Methods for Physics, UC Irvine

**Physics Tutor** *2019*
UC Irvine

**Physics Tutor** *2017 – 2019*
UC Davis Physics Club, UC Davis